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Optimizing valve actuator parameters to enhance control valve performance

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Abstract

The market demands for higher quality products have increased the need for reduced process variation. One factor in the quality of the final end product is the improvement of the control loop performance. A critical component in the loop is the final control element, the control valve package. Optimized actuator parameters play a vital role in the dynamic performance of the control valve. This paper reviews the pneumatic actuator and positioner parameters that effect the control package performance. This is done through the use of a control valve package computer model to assess the dynamic performance. The attributes of spring return versus double acting actuators are illustrated. The effects of supply pressure, step size, load margin, and flow, actuator volume and design style are investigated through the use of mathematical simulations of control valve dynamic performance. Copyright © 1996 Elsevier Science Ltd.

Keywords: Control valves; Actuators; Positioners

1. Introduction

A computer model has been developed [1] to assess the dynamic performance of a control valve, actuator and positioner (control package) assembly. Through this model, a parametric analysis is made of load margin, supply pressure and flow, actuator volume and design style.

2. Background

In the modeling of the control valve, actuator and positioner, the component kinematics, geometry, fluid flow and friction must be considered. The model for a double acting actuator may be viewed schematically as shown in Fig. 1.

The action of the three major components: valve, actuator and positioner are integral to the performance of the control package. Each component is modeled to allow analysis of the interaction of parameters on the performance of the integrated control valve package. The mathematical model of the control package was described in [1]. Here, the kinematic and fluid dynamic properties of each component were combined to give an overall simulation of the control valve package.

3. Friction modeling

In order to model any of the components, the mechanical friction must be addressed. The mathe-

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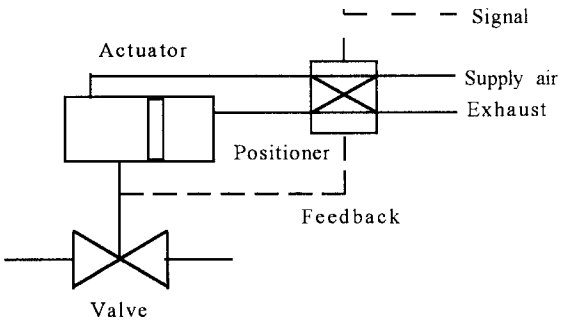


Fig. 1. Control valve package schematic.

matical model includes the formulation of the dynamic as well as the static coefficients of friction (Fig. 2).

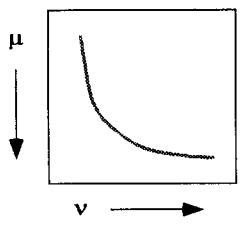
4. Valve / actuator transform modeling

Linear valves have a simple, direct coupling of the valve load to the actuator. However, rotary valves can have a variety of linkages to convert the linear piston or diaphragm force to rotary motion and torque. For example, some double acting and spring return profiles are shown in Fig. 3.

As with a linear valve, the spring constant alters the load output of the actuator air stroke. However, for rotary valves, the valve motion and forces must be transformed into actuator rotary coordinates. The mathematical model includes position dependent coefficient tables to support this transform.

5. Positioner modeling

The positioner can have a significant influence on the control package dynamics. Ideally, the flow gain



$$\mu = \mu_d + (\mu_s - \mu_d)e^{-\gamma v^{\delta}}$$

μ	- coefficient of friction
d	- dynamic
s	- static
γ	- stick-slip parameter
δ	- material parameter
v	-velocity

Fig. 2. Dynamic coefficient of friction.

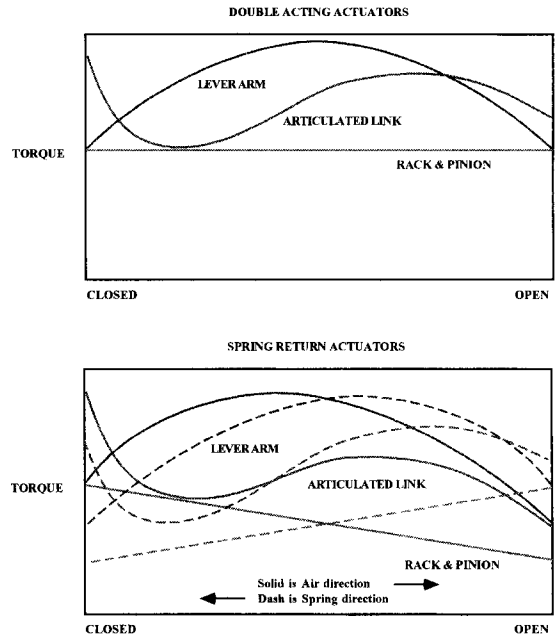


Fig. 3. Linear to rotary force transforms — torque versus valve position.

and pressure gain [2] should be linear within the normal span of the error signal. In Fig. 4a, the flow gain versus signal error is shown for an example positioner. In normal control operation, error signals do not exceed 10% of full span so, the flow gain is linear in the normal signal error range (perhaps 0.5–10% of full span). Similarly, Fig. 4b shows a graph of the pressure gain relative to the error signal. The gain is constant above very small error signals and throughout the normal control range.

6. Step response

There are many measures of control valve performance including non dynamic measures such as

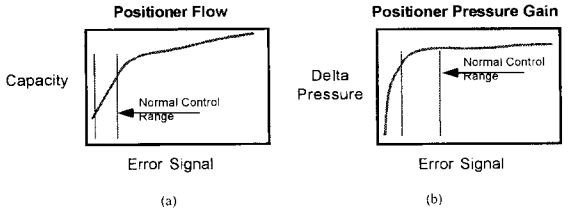
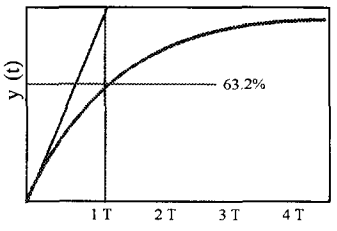


Fig. 4. Positioner flow and pressure gain profiles.



$$T\frac{d(y(t))}{dt} + y(t) = Gi$$

$$y(t) = G\left[1 - e^{\frac{-t}{T}}\right]$$

Fig. 5. Output versus input–first order.

linearity, repeatability, hysteresis and deadband. See Ref. [2] for definitions and details on these factors. One key measure of dynamic performance is the step response of the control valve package. This is the measure used here to evaluate the relative performance of control valve packages. Normally, the control valve shaft is monitored as the step signal is sent to the positioner. This step is normally in the range of 2–10% of full span. The time constant, T (Tau), is the time required for the output to move to 63% of the full output response to the step change in input. Fig. 5 illustrates the response $y(t)$ of a first order system to a step change in input where T is the time constant, G is the gain and t is the time.

In addition, the time for the output to start to change in response to the input step is the dead time, *td*. During this dead time the air pressure in the actuator must build to the unbalanced force necessary to move the valve. These two times, *td* and T are used herein to compare the performance of control valve packages with parametric variations.

7. Computer program output

The computer program used for this parametric analysis calculates the dead time and time constant for the control package in response to a step change in input signal. This program is based on the mathematical model described earlier and in Ref. [1]. Fig. 6 shows an example of the output. The valve travel is shown relative to the signal input and the pressure(s) in the actuator are both shown relative to time. The actuator cylinder pressures create the dif-

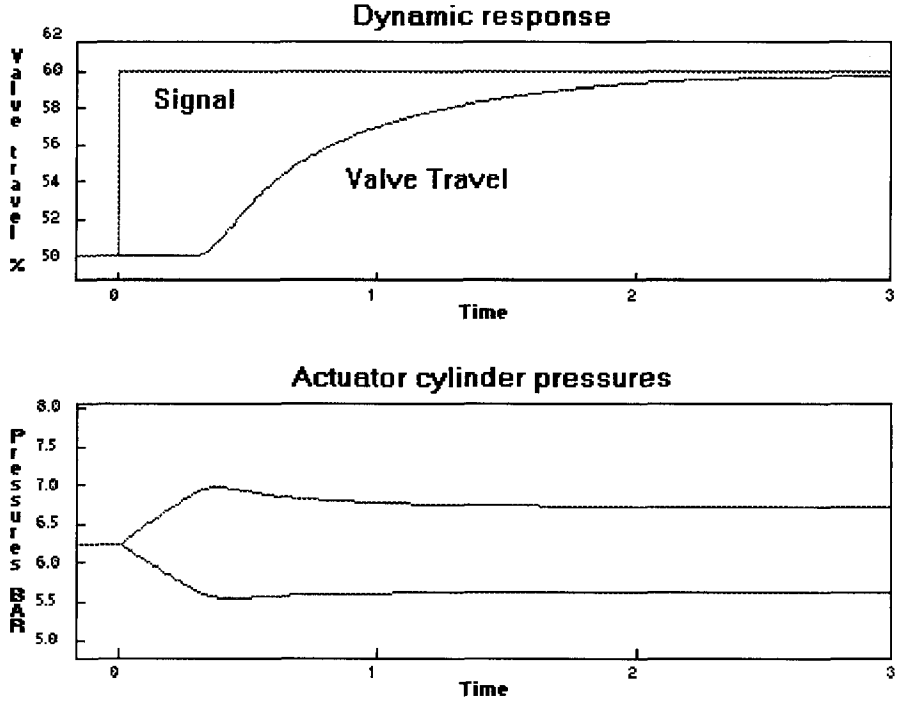


Fig. 6. Theoretical position change versus step change input — time is in sec.

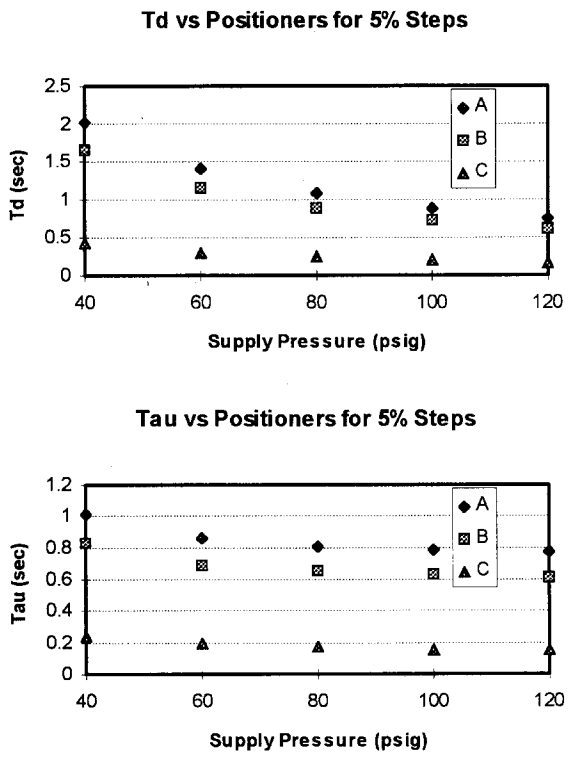


Fig. 7. Effects of flow gain and supply pressure variation.

ferential load to move the valve. Once in position, the differential will re-converge to the average pressure.

8. Supply pressure

The supply pressure delivered to the actuator would seem to influence the control package speed of a response by providing a higher differential to increase the flow rate and by raising the load margin of the actuator over the valve. A series of runs were made with the same valve and actuator package. The dead time and time constant were recorded for step inputs of 2, 5, and 10% of full span for a double acting actuator. The supply pressure parameter was varied: 40, 60, 80, 100, 120 psi and the positioner flow capacity was varied in three levels: A, B, C (lowest to highest). Fig. 7 illustrates the dead times and time constants for the 5% step size.

In both cases, the dead times and time constants are reduced as the supply pressure is increased. This reduction is roughly by 65% for *td* and 25% for tau. These time reductions, due to an increase in air supply pressure, are consistent for a given positioner and (not shown) roughly consistent for various step sizes. In short, the time reductions are approximately consistent for a given flow capacity.

Also, the dead time is consistently greater than the time constant for this control package. As shown in Fig. 7, before the valve can change position the actuator must create an unbalanced load to overcome the valve resisting load. Once moving, the unbalanced force is maintained with the assistance of the momentum of mechanical parts.

It can be concluded that increases of supply pressures significantly increase the speed of response of a control valve package.

9. Step signal size

Next, the influence of the magnitude of the step change of the input signal is investigated. A series of

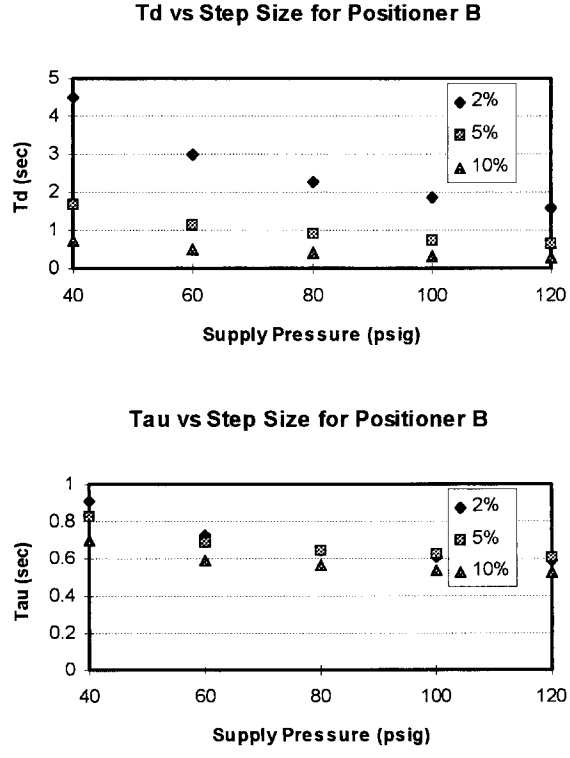


Fig. 8. Effects of step signal size and supply pressure variation.

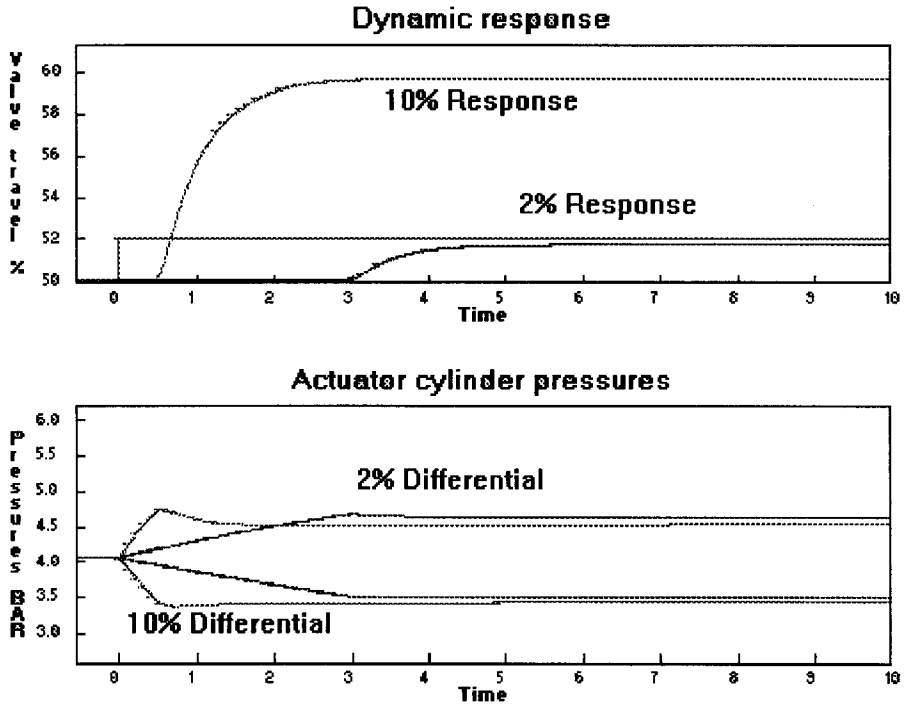


Fig. 9. Dead time versus step size: pressure build up.

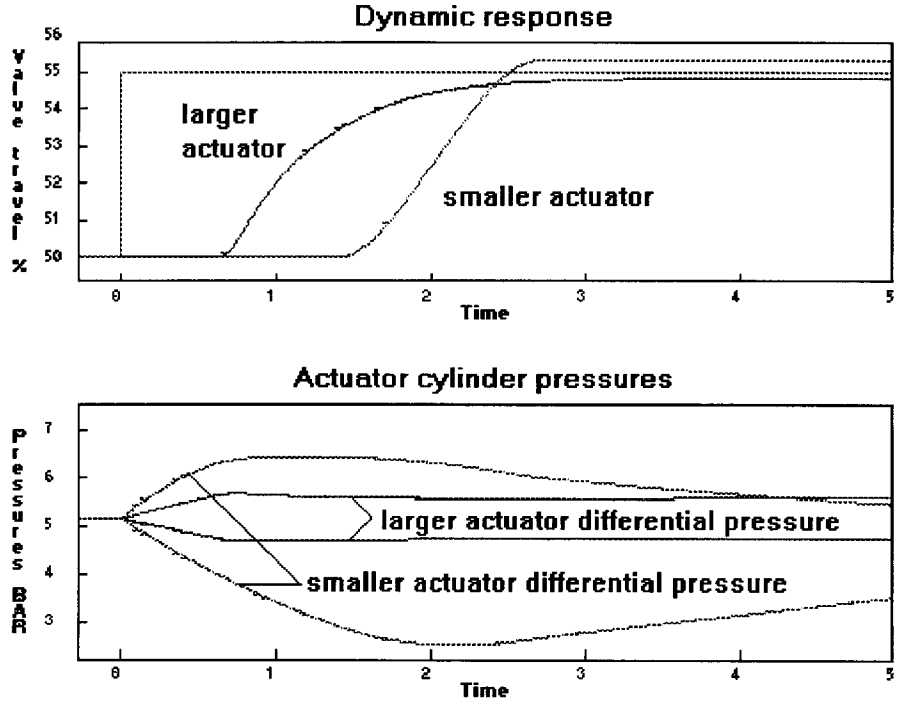


Fig. 10. Torque (force) margin decreases response time.

runs were made where the dead time and time constant were recorded for various positioner flow capacities in three levels: A, B, C (low to high capacity) with a double acting actuator. The supply pressure parameter was varied: 40, 60, 80, 100, 120 psi and the input step size was varied through 2, 5, 10% of full span. Fig. 8 illustrates the dead times and time constants for the size B positioner.

When the input step size is varied, the dead time in Fig. 8 decreases with the step size. The dead time for the 5% step is 60% of that for the 2% step. Similarly, the dead time for the 10% step is 80%. The impact is consistent for supply pressure variation and (not shown) for positioner size. This result confirms the model of the linear proportional gain for the positioner discussed earlier.

Step size has a far smaller impact on the time constant. In fact, the time constant of the control package is relatively constant with the variation in step size as shown in Fig. 9. It seems that the kinematics of the control package effects the load balance of the system once the valve and actuator elements are moving. As shown in Fig. 9, the larger step change signal creates a greater pressure unbalance in the actuator. This coupled with the valve/actuator kinematics allows the control package to make a larger step change in position in nearly the same time constant as the smaller step change. The total time difference is mainly that caused by dead time.

10. Actuator sizing-load margin and volume

The sizing of the actuator can have a significant impact on the performance on the control valve package. Fig. 10 shows the relative step response of a control package with two different size actuators. The undersized actuator shows a longer dead time but a shorter time constant than the larger actuator. The reason is seen in the actuator pressure differential curve. The smaller actuator needs a higher differential pressure to create the load unbalance to move the valve. However, since the actuator is smaller, it is easier to keep this differential pressure as the control valve package strokes due to the smaller stroke volume.

11. Spring return actuators

Spring return actuators. in some ways, react differently to a variation in parameters than double acting actuators. First, the spring return actuator unbalances the spring using air pressure to create load unbalance to move the control valve. Once in position, the spring is counterbalanced with the appropriate air pressure to prevent motion. Higher supply pressures are not translated into higher pressures in the actuator; therefore, the stroke speed created by the spring is not effected by a variation in supply

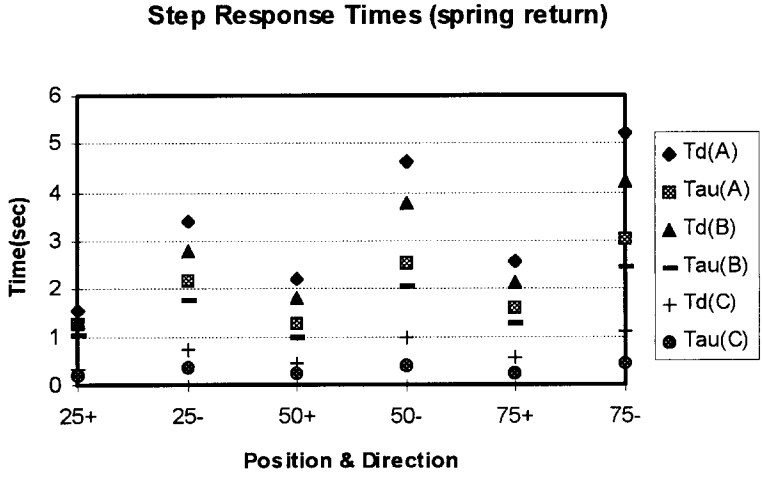


Fig. 11. Effects of direction and position in spring return actuators.

pressure. The stroke speeds during the air stroke are effected somewhat, since the differential across the positioner may be increased similar to that developed with the double acting actuator discussed earlier. It follows that the step size would impact the dynamic performance of the spring return actuator similar to that of the double acting actuator. And, actuator sizing has similar benefits to the double acting as well.

The dead time and time constants were investigated for a spring return actuator at various positions through the valve stroke. The balance between the spring and the air changes throughout this stroke as a result of the spring constant. The results are shown for an example 5% step signal change in Fig. 11.

Both the initial position, 25, 50, 75%, and direction, + is air stroke and − is spring stroke, are shown for a variation in positioner size. Two features are noted. First, the difference between the step response in the spring direction versus the air directions is apparent. The relationship can vary with the control valve package. Second, the effects of the spring constant are shown as the valve is moved toward the open position where the spring force and opposing air pressure increase with the percent opening.

12. Conclusions

The control valve package is a complicated dynamic system where the performance can be significantly effected by the variation of certain parameters. The selection and use of valve, actuator and positioner may be assisted by guidelines uncovered here. Clearly, higher supply pressure is the single greatest benefit in the speed of response of a control valve package using a double acting actuator. However, the interaction of kinematics and fluid dynamics is, at times, not obvious. The more reliable prediction of control valve package performance may result in the use of a mathematical model of the valve, actuator and positioner package.

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